WORKED SOLUTION

STEP III 2002, Question 11

Mechanics • Collision

STEP III 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q11

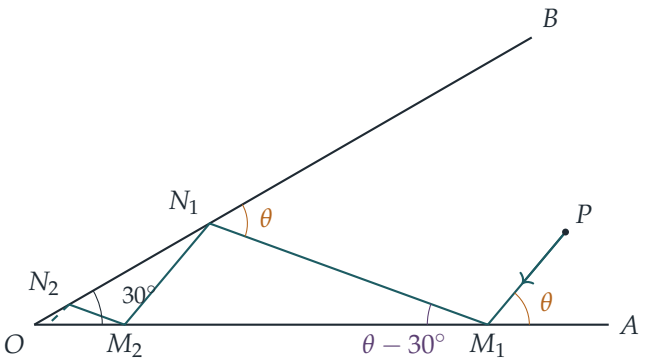
This question has a full guided treatment in *Beyond STEP*, Section 10.5, *Infinitely Many Bounces in Finite Time*: https://beyondstep.keithmcnulty.org/ch10/zeno-wedge.html

The key idea

At a smooth wall the impulse is perpendicular to the wall. So the velocity component along the wall survives unchanged and only the perpendicular component is multiplied by e. Combine this with the angles of the triangle OM_1N_1 and the first part reduces to one trigonometric equation. For the second part, the motion repeats itself on a smaller scale after each bounce. The legs get shorter, but the particle also slows down, so we have to add up the *times*, not the distances. That sum is a geometric series.

Solution

We model the particle as a point moving on a smooth horizontal surface, so between impacts it moves in a straight line at constant speed. The walls are smooth, and each impact obeys Newton's experimental law with coefficient e.



Finding e and v_1/u_1

Since angle $PM_1A = \theta$, the particle reaches M_1 travelling towards O, at angle θ to the wall. Just before impact, its velocity components are

$u_1\cos\theta$ along the wall (towards O), $\qquad$ $u_1\sin\theta$ perpendicular to the wall (into it).

The wall is smooth, so the impulse on the particle is perpendicular to OA. Hence:

along the wall: $u_1\cos\theta$ (unchanged),
perpendicular: $e\,u_1\sin\theta$ (reversed and multiplied by e).

So the particle leaves M_1 still moving towards O, at an angle φ to the wall, where

$$\tan\varphi = \frac{e\,u_1\sin\theta}{u_1\cos\theta} = e\tan\theta, \qquad v_1\cos\varphi = u_1\cos\theta. \tag{$*$}$$

Next we find φ from the geometry of triangle OM_1N_1. The points O, N_1, B lie on one line, and angle $M_1N_1B = \theta$, so angle $ON_1M_1 = 180^\circ - \theta$. The angle at O is 30°, so

$$\varphi = \angle OM_1N_1 = 180^\circ - 30^\circ - (180^\circ - \theta) = \theta - 30^\circ.$$

For φ to be positive we need $\theta > 30^\circ$, which this configuration implicitly assumes. Then $(*)$ gives

$$\boxed{e = \frac{\tan(\theta - 30^\circ)}{\tan\theta}}$$

and

$$\boxed{\frac{v_1}{u_1} = \frac{\cos\theta}{\cos(\theta - 30^\circ)} = \frac{2}{\sqrt{3} + \tan\theta}},$$

where the last form comes from expanding $\cos(\theta - 30^\circ) = \frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta$. If you prefer, write $e = \dfrac{\sqrt{3}\tan\theta - 1}{\tan\theta\,(\sqrt{3} + \tan\theta)}$. Since $0 < \theta - 30^\circ < \theta < 90^\circ$, we get $0 < e < 1$, as the question requires.

Reaching O in a finite time

At N_1 the particle arrives travelling towards O, at angle θ to the wall OB (because angle $M_1N_1B = \theta$). This is exactly the situation at M_1, with the roles of the walls swapped. So the same calculation applies: it rebounds at angle $\theta - 30^\circ$ to N_1O, and its speed is multiplied by

$$r = \frac{\cos\theta}{\cos(\theta - 30^\circ)}.$$

Ask yourself Does the particle arrive at M_2 at angle θ again, so that the pattern really repeats?

Yes. In triangle ON_1M_2 the angle at O is 30° and the angle at N_1 is $\theta - 30^\circ$. So angle $OM_2N_1 = 180^\circ - \theta$, which means angle $N_1M_2A = \theta$. By induction, every impact happens at angle θ to the wall. Every leg leaves its wall at angle $\theta - 30^\circ$. And every triangle OM_nN_n, ON_nM_{n+1} has angles 30°, $\theta - 30^\circ$, $180^\circ - \theta$, so they are all similar.

Lengths. By the sine rule in triangle OM_1N_1,

$$\frac{ON_1}{\sin(\theta - 30^\circ)} = \frac{OM_1}{\sin(180^\circ - \theta)} = \frac{M_1N_1}{\sin 30^\circ}.$$

So each new triangle is the previous one scaled by

$$k = \frac{ON_1}{OM_1} = \frac{\sin(\theta - 30^\circ)}{\sin\theta}.$$

Call the legs M_1N_1, N_1M_2, $M_2N_2, \ldots$ legs $1, 2, 3, \ldots$. Leg n has length $\ell_1 k^{n-1}$, where $\ell_1 = M_1N_1 = \dfrac{OM_1}{2\sin\theta}$.

Speeds. Leg n is travelled at constant speed $v_1 r^{n-1}$.

Times. Leg n therefore takes time

$$\tau_n = \frac{\ell_1 k^{n-1}}{v_1 r^{n-1}} = \frac{\ell_1}{v_1} q^{n-1}, \qquad q = \frac{k}{r} = \frac{\sin(\theta - 30^\circ)\cos(\theta - 30^\circ)}{\sin\theta\cos\theta} = \frac{\sin(2\theta - 60^\circ)}{\sin 2\theta}.$$

Ask yourself The legs shrink geometrically, but the particle is slowing down too. Why does that not settle the matter?

The times are what we add up, and the time ratio q compares the shrinking of the legs (k) with the slowing of the particle (r). We need $q < 1$. Now suppose $30^\circ < \theta < 60^\circ$. Then

$$0^\circ < 2\theta - 60^\circ < 60^\circ \;\Rightarrow\; \sin(2\theta - 60^\circ) < \tfrac{\sqrt{3}}{2}, \qquad 60^\circ < 2\theta < 120^\circ \;\Rightarrow\; \sin 2\theta > \tfrac{\sqrt{3}}{2},$$

so $0 < q < 1$. The total time is then a convergent geometric series:

$$T = \frac{PM_1}{u_1} + \sum_{n=1}^{\infty} \frac{\ell_1}{v_1} q^{n-1} = \frac{PM_1}{u_1} + \frac{\ell_1}{v_1(1-q)}.$$

This is finite. To simplify, use $\sin 2\theta - \sin(2\theta - 60^\circ) = 2\cos(2\theta - 30^\circ)\sin 30^\circ$, which gives $1 - q = \dfrac{\cos(2\theta - 30^\circ)}{\sin 2\theta}$. Then the time from M_1 onwards is

$$\frac{\ell_1}{v_1(1-q)} = \frac{OM_1\cos(\theta - 30^\circ)}{u_1\cos(2\theta - 30^\circ)},$$

which is positive and finite for $30^\circ < \theta < 60^\circ$.

Finally, the particle really does arrive at O. After the nth impact its distance from O is $OM_1k^n \to 0$, because $0 < k < 1$. During a leg, its distance from O is at most the larger of the distances of the leg's two ends. So as $t \to T$ the particle's distance from O tends to 0: it reaches O at the finite time T, after infinitely many impacts. ■

Checks

- At $\theta = 45°$: $e = \tan 15° = 2 - \sqrt{3} \approx 0.268$, $v_1/u_1 = \sqrt{3} - 1 \approx 0.732$, and $q = \sin 30° / \sin 90° = \frac{1}{2}$, so each leg takes half as long as the one before. With $OM_1 = u_1 = 1$, the time from M_1 to O is $\cos 15° / \cos 60° = 2\cos 15° \approx 1.932$. A step-by-step simulation of the impacts (sixty bounces) gives 1.931852, in agreement.
- At $\theta = 60°$: $q = 1$, so every leg takes the same time and the total is infinite. For $\theta > 60°$, $q > 1$. So $60°$ is exactly the borderline, and the formula for T blows up there through $\cos(2\theta - 30°) = \cos 90° = 0$.
- As $\theta \to 30°^+$, $e \to 0$: the particle barely bounces and slides almost straight to O, as you would expect.

Common slips

- Getting the triangle wrong: angle M_1N_1B is the *exterior* angle at N_1, so the interior angle is $180° - \theta$ and the rebound angle at M_1 is $\theta - 30°$, not $\theta + 30°$.
- Arguing that the particle reaches O just because the distances $OM_n \to 0$. Infinitely many legs need a convergent series of *times*, and the particle slows down at each impact.
- Reversing the tangential component at impact. On a smooth wall only the normal component changes.

Beyond STEP

Infinitely many impacts in a finite time is called *Zeno behaviour*. The familiar example is a ball bouncing on the floor with $e < 1$, and here the wedge plays the role of the floor. Mathematicians studying 'hybrid' systems, which mix smooth motion with sudden jumps, must decide what happens after the Zeno time. In granular materials the same effect, called *inelastic collapse*, causes real trouble in simulations. For collisions and impulse treated carefully at first-year level, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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